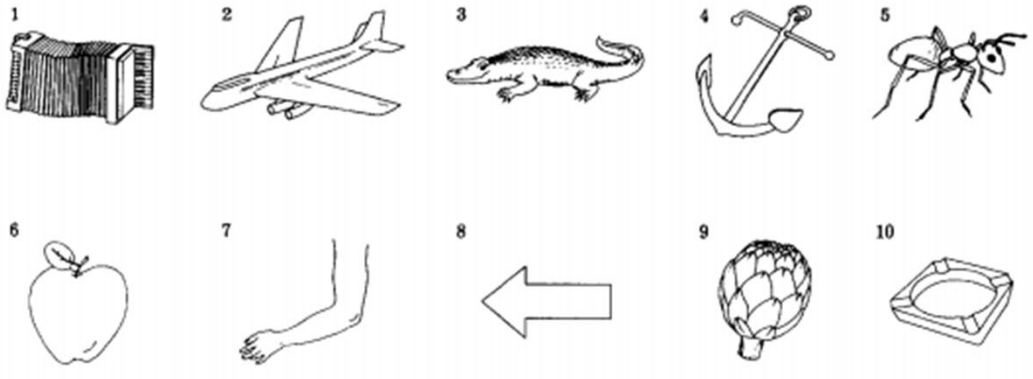
Experimental Design

A classic cautionary tale:
The frequency effect, and its confound with length

Oldfield and Wingfield (1965) asked people to name pictures of objects.

They found a log-linear relationship between frequency and response time.



Note that frequency is related to word length, however, even in this tiny sample.

Let's look at this quantitatively, using a web-based tool…

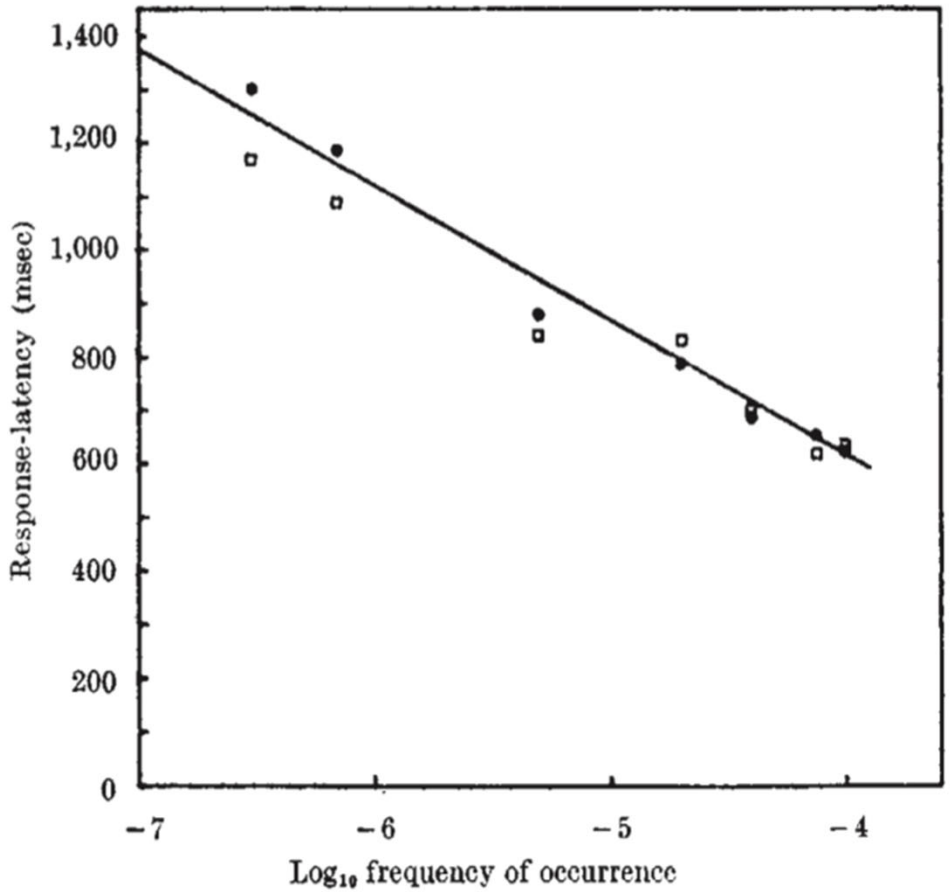
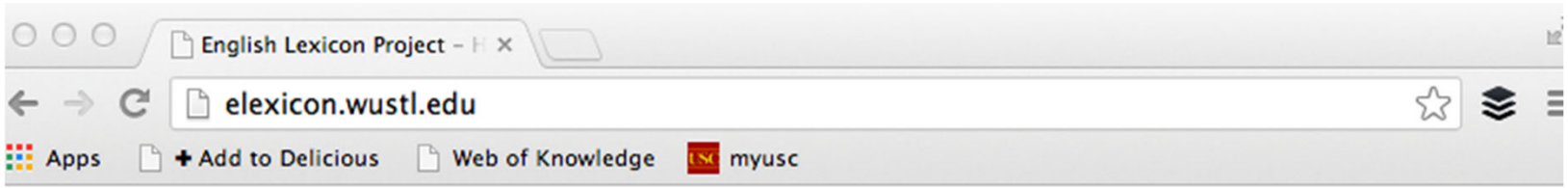


Fig. 1. Response-latency in naming objects versus $\log_{10}$ frequency of occurrence of the name in English. ●, Picture set 1; □, picture set 2

Oldfield, R. C. & Wingfield, A. (1964) The time it takes to name and object. *Nature, 202,* 1031-1032

…these line drawings are from Snodgrass & Vanderwart (1980)…



Washington University in St.Louis

English Lexicon Project Web Site

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Data Entry

Site Administration

Welcome to the English Lexicon Project Homepage.

Overview

The English Lexicon Project (supported by the National Science Foundation) affords access to a large set of lexical characteristics, along with behavioral data from visual lexical decision and naming studies of 40,481 words and 40,481 nonwords.

The naming and lexical decision data are currently being collected from six testing Universities. To date, we have collected 2,752,698 reaction time measurements from 816 subjects in the lexical decision experiment. We have also collected 1,125,880 experimental measurements from 444 subjects in the naming experiment.

Database News and Update ~ 11/10/2009 ~ Click Here!

There are four main functions of this Web Site:

- **Generate Lists of Words with Specific Lexical Characteristics**
- **Generate Lists of Nonwords with Specific Lexical Characteristics**
- **Access Trial-Level Lexical Decision Data**
- **Access Trial-Level Naming Data**

Click Here for User's Guide...

Click Here for Levenshtein Distance Paper...

Click Here for Brysbaert – New Behavior Research Methods...

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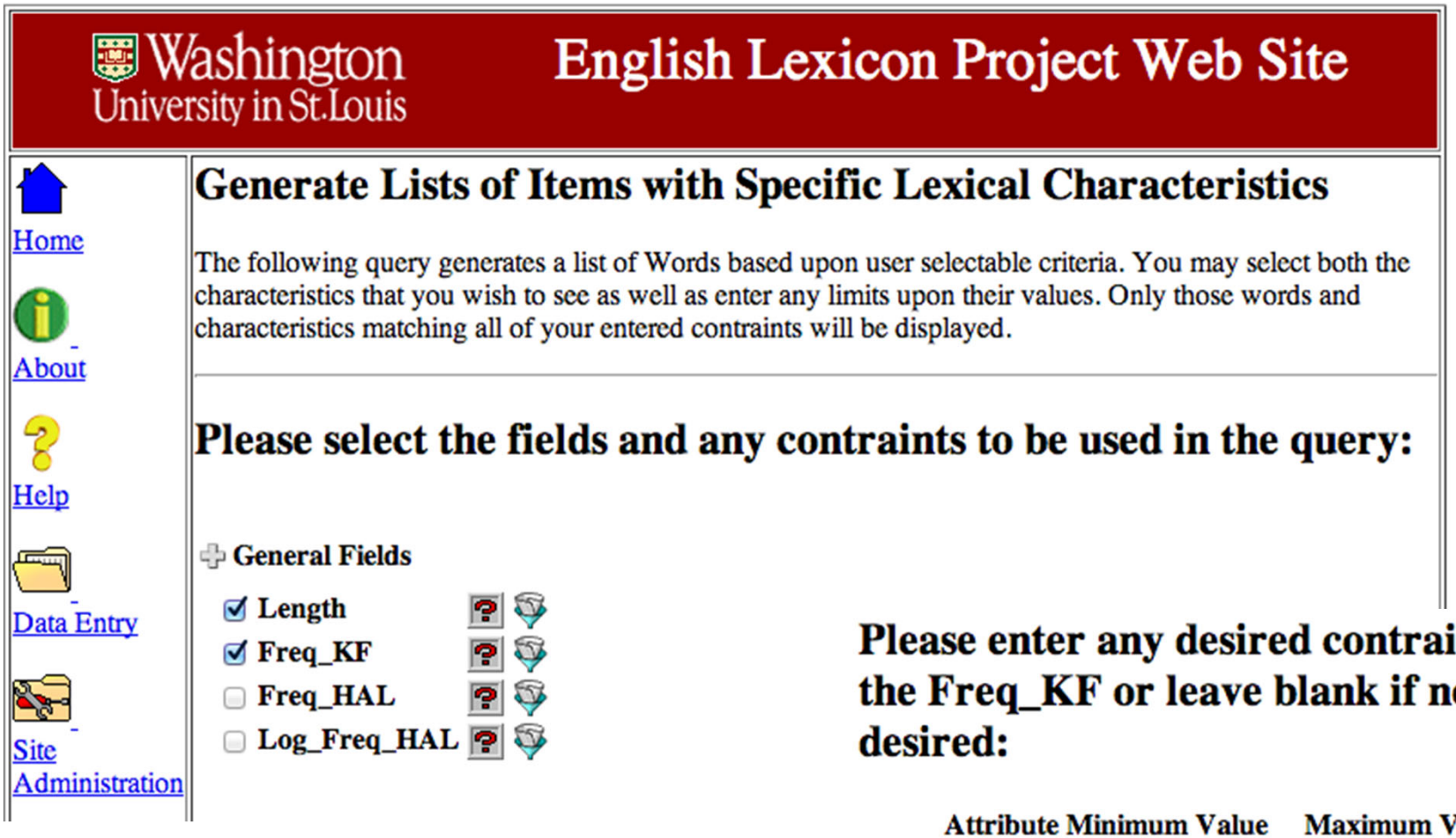
Site Administration

Word Queries - Access the properties of Words contained in the ELP Database

There are two word based queries available on this Web Site:

- **Generate Lists of Items with Specific Lexical Characteristics**
- **Submit Lists of Items to generate Lexical Characteristics**

Last updated on 11/10/2009.
Webmaster: Bjorn Loftis: Email, Web Site
© copyright 2001-2009 The Cognitive Psychology Laboratory, Washington University
Unauthorized Use Strictly Prohibited: *Word lists generated from this website are available for non-commercial research purposes only and may not be used in the development of speech technology.*

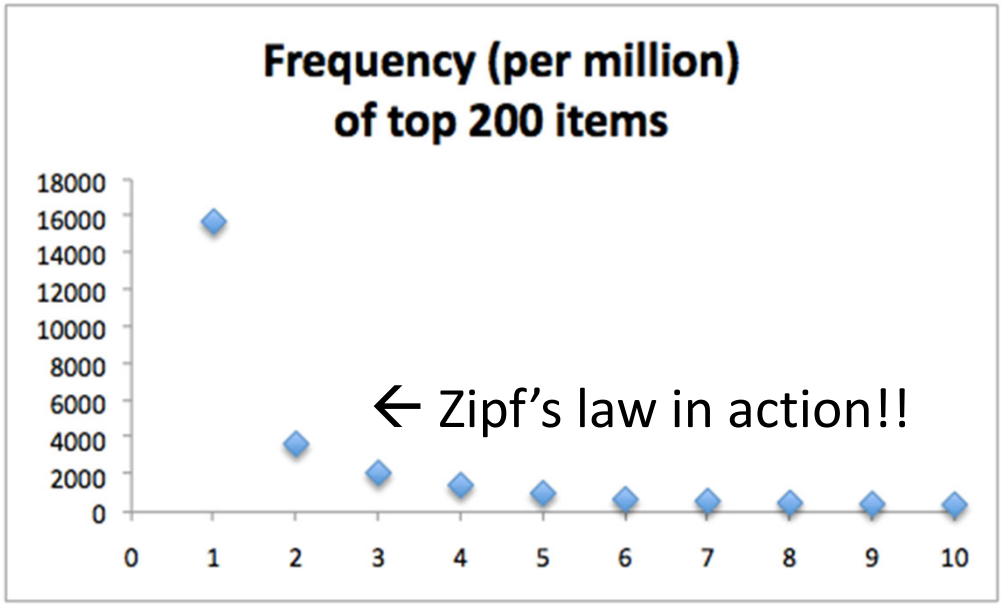

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Generate Lists of Items with Specific Lexical Characteristics
The following query generates a list of Words based upon user selectable criteria. You may select both the characteristics that you wish to see as well as enter any limits upon their values. Only those words and characteristics matching all of your entered contraints will be displayed.
Please select the fields and any contraints to be used in the query:
General Fields
Length
Freq_KF
Freq_HAL
Log_Freq_HAL
Please enter any desired contraints upon the Freq_KF or leave blank if non are desired:
Attribute Minimum Value Maximum Value
Freq_KF 200 100000
Submit Constraint

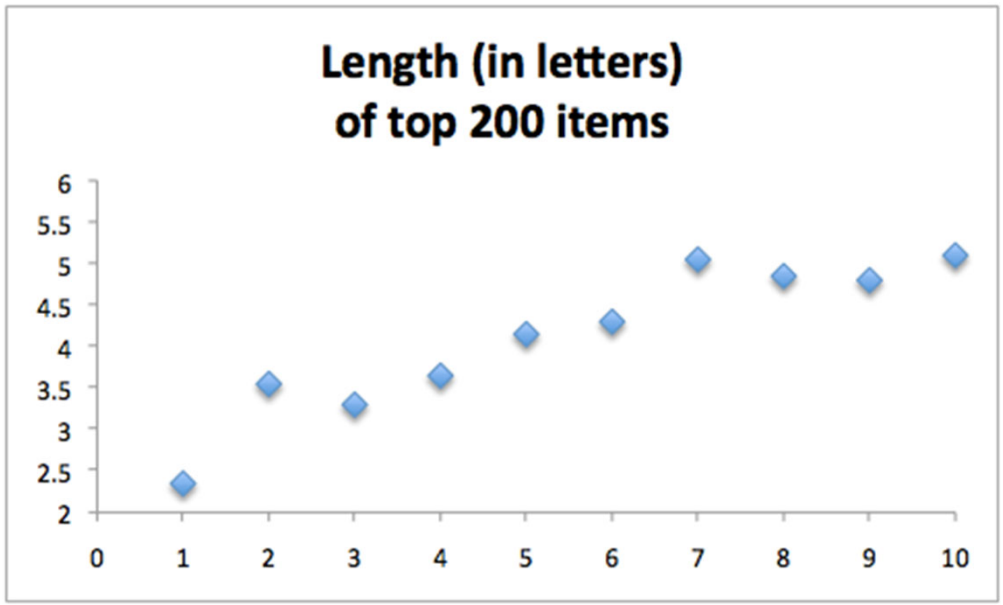
The 20 most frequent words in English

Word	Length	Freq_KF
the	3	69,971
of	2	36,411
and	3	28,852
to	2	26,149
a	1	23,237
in	2	21,341
that	4	10,595
is	2	10,099
was	3	9,816
he	2	9,543
for	3	9,489
it	2	8,756
with	4	7,289
as	2	7,250
his	3	6,997
on	2	6,742
be	2	6,377
at	2	5,378
by	2	5,305

20 more high-frequency words

century	7	207
Washington	10	206
usually	7	206
north	5	206
therefore	9	205
plan	4	205
leave	5	205
top	3	204
sound	5	204
million	7	204
evidence	8	204
black	5	203
strong	6	202
hard	4	202
various	7	201
value	5	200
type	4	200
surface	7	200
says	4	200
play	4	200



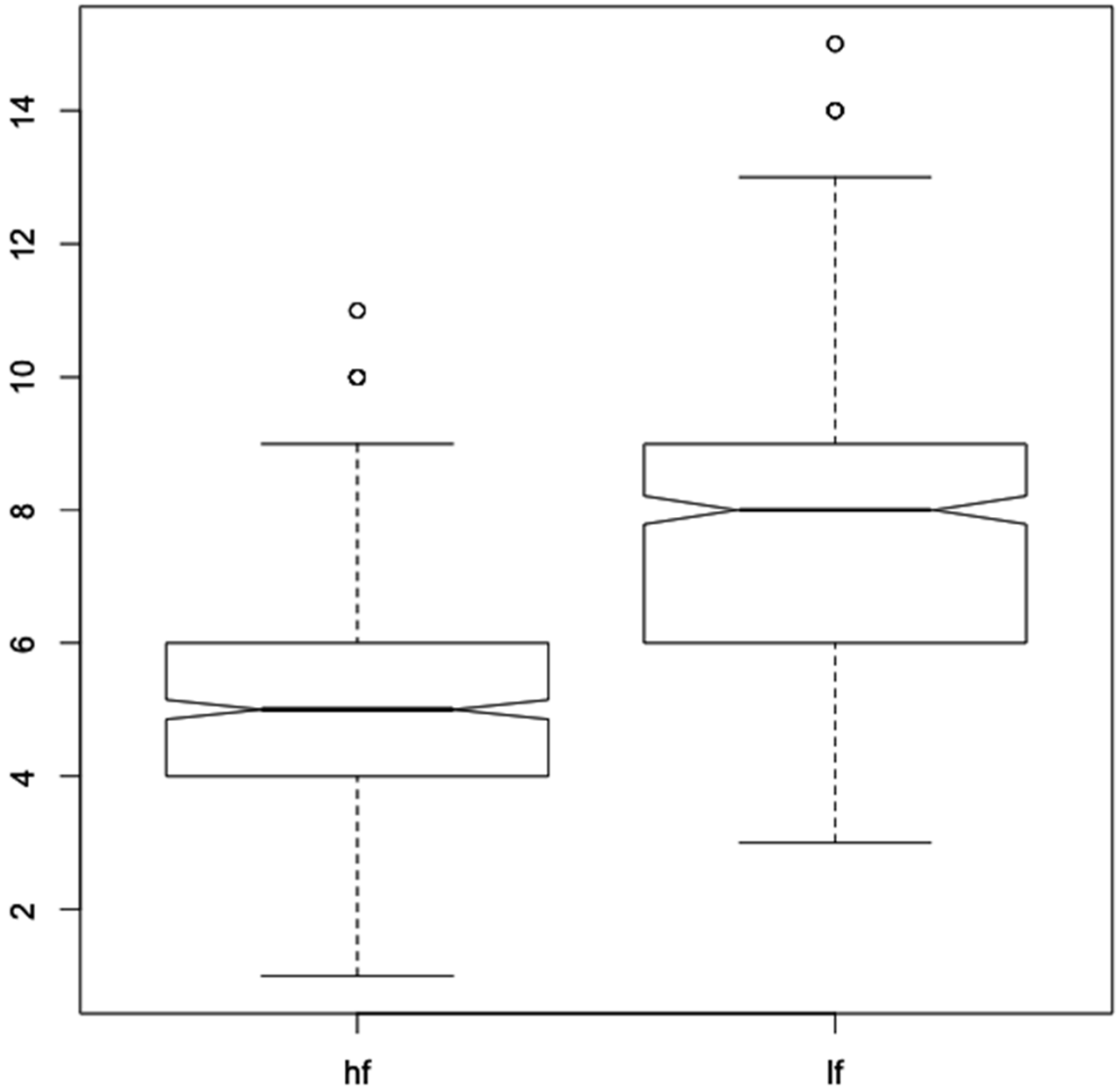


What I did here was take the 200 most frequent words in an old corpus, and put them into ten "bins" of twenty words each.

Notice:

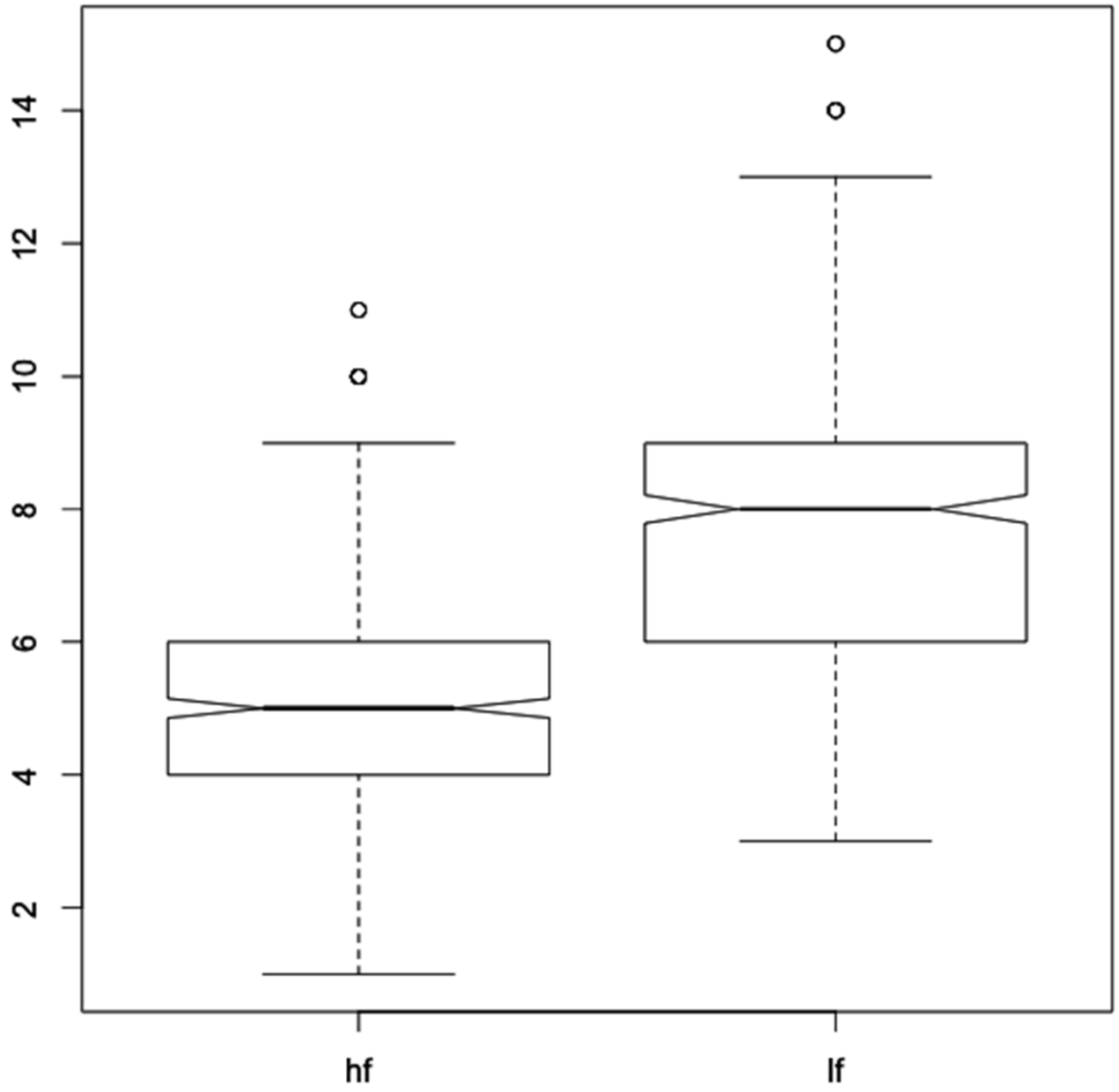
1. The mean frequency drops off very dramatically from the most frequent items to the next most frequent, etc.

2. Length is inversely related to frequency.



Boxplots show distributions of the lengths of the 467 most frequent words (high frequency = “HF”) and 467 low frequency (“LF”) words selected at random.

Fewer than 25% of HF words are as long as the average (median) LF word.



Problem: HF words are just generally shorter than LF words, so this will always happen if you select LF words at random.

Solution: *don't select LF words at random!*

If you select long-ish HF words and short-ish LF words, you can match on length while manipulating frequency…

Oldfield and Wingfield eventually addressed this by matching their high- and low-frequency words for length.

They found a robust frequency effect, so they concluded that frequency effects are not merely length effects in disguise.

Really rather a lot of psycholinguistics can be characterized as an attempt to demonstrate that some observed effect is not just an effect of some other confounding variable…

“Making up materials is a confounded nuisance”*

Table 1
Summary Statistics for the Items Used in the Experiment (Mean Values are Provided)

Variable	Regular		Irregular	
	F > *E*	*E* > *F*	*F* > *E*	*E* > *F*
Frequency	16.8	10.8	13.5	11.7
Familiarity	6.25	6.41	6.41	6.44
Imageability	5.07	4.95	4.99	5.10
Length	4.5	4.5	4.7	4.9
Bigram-frequency	4,997	5,312	5,730	5,767
Coltheart’s *N*	7.1	7.7	6.4	5.2
Mean summed frequency–friends	449.1	184.0	745.8	140.7
Mean summed frequency–enemies	125.5	873.3	175.5	569.0
Conditional probability–body	.84	.41	.70	.27
Conditional probability–grapheme	.57	.62	.12	.17

A pretty beautiful matching job by Cortese & Simpson (2000)

Careful matching on a wide range of dimensions is an important part of good stimulus design for an experiment.

This can get very sophisticated.

One concern with this very fine-grained stimulus matching is that we may select items that are somehow “odd,” or for which the statistics are inappropriate.

Remember that the confound between frequency and length (for example) is a fact about English (and most languages). Picking items to avoid this confound -- even if done well -- creates a somewhat “artificial” stimulus environment.

Or we might end up with a very small number of items (the study I took this table from had *ten* words per cell, which is indeed small for a word naming experiment).

*Cutler, A. (1981) Making up materials is a confounded nuisance, or: Will we be able to run any psycholinguistic experiments at all in 1990? *Cognition, 10,* 65-70.

Task Design



Most interesting phenomena in psycholinguistics happen below the level of conscious awareness.

Before you started this class, you would probably not be thinking about word frequency or length if I asked you to name a bunch of line drawings on a screen.

So, psycholinguists don't have to be as crafty about experimental design as people studying, say, racial stereotyping, or altruism.

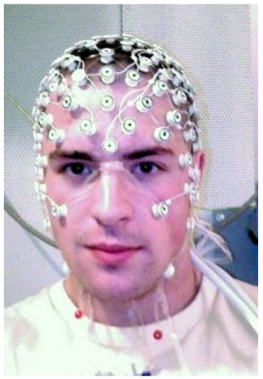
Indeed, in norming studies, we routinely just ask people to rate words on some dimension, and just hope for the best.

(But remember our example in class did have "fillers.")

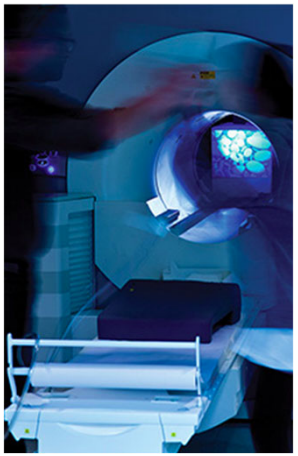
Task Design



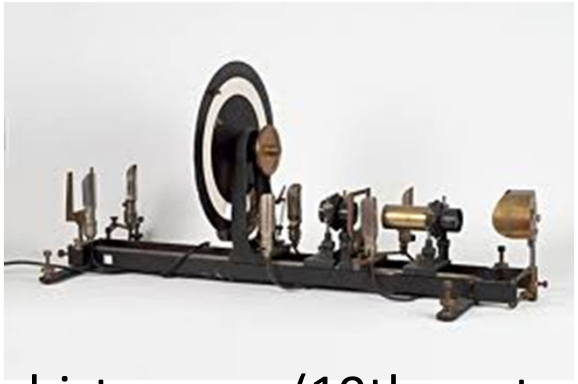
eye-tracking



eeg



FMRI



tachistoscope (19th century)

Many tasks take advantage of our ability to measure things that participants don't exercise conscious control over.

Eye-movements can be very revealing, as we'll see when we discuss speech production studies later today.

EEG and FMRI both measure brain activity.

Often this is most interesting not for what it tells us about the brain per se, but because brain activity is a dependent variable that participants don't (usually) control directly.

So, for example, if we observe different patterns of brain response to different kinds of stimuli *even in the absence of an overt task,* it suggests that those stimuli are different to the participant.

Of course, reaction time and accuracy are also unlikely to be under *direct* control in most experiments...

Some example experiments...

Over the next three or four sessions, we'll be talking about a handful of specific research studies on specific topics such as:

- perception and categorization of speech sounds
- spoken word recognition (and how/whether it interacts with speech sounds)
- sentence comprehension

These are huge areas of research that don't get much coverage in the textbook.

Today we'll spend some time with the Belke et al. paper discussed in the text, since it has some nice design features, and is about speech production, which you're all experts on by now…

THE QUARTERLY JOURNAL OF EXPERIMENTAL PSYCHOLOGY
2005, 58A (4), 667–692

Psychology Press
Taylor & Francis Group

Refractory effects in picture naming as assessed in a semantic blocking paradigm

Eva Belke and Antje S. Meyer
The University of Birmingham, Birmingham, UK

Markus F. Damian
University of Bristol, Bristol, UK

In the cyclic semantic blocking paradigm participants repeatedly name sets of objects with semantically related names (*homogeneous sets*) or unrelated names (*heterogeneous sets*). The naming latencies are typically longer in related than in unrelated sets. In Experiment 1 we replicated this semantic blocking effect and demonstrated that the effect only arose after all objects of a set had been shown and named once. In Experiment 2, the objects of a set were presented simultaneously (instead of on successive trials). Evidence for semantic blocking was found in the naming latencies and in the gaze durations for the objects, which were longer in homogeneous than in heterogeneous sets. For the gaze-to-speech lag between the offset of gaze on an object and the onset of the articulation of its name, a repetition priming effect was obtained but no blocking effect. Experiment 3 showed that the blocking effect for speech onset latencies generalized to new, previously unnamed lexical items. We propose that the blocking effect is due to refractory behaviour in the semantic system.

Belke et al.

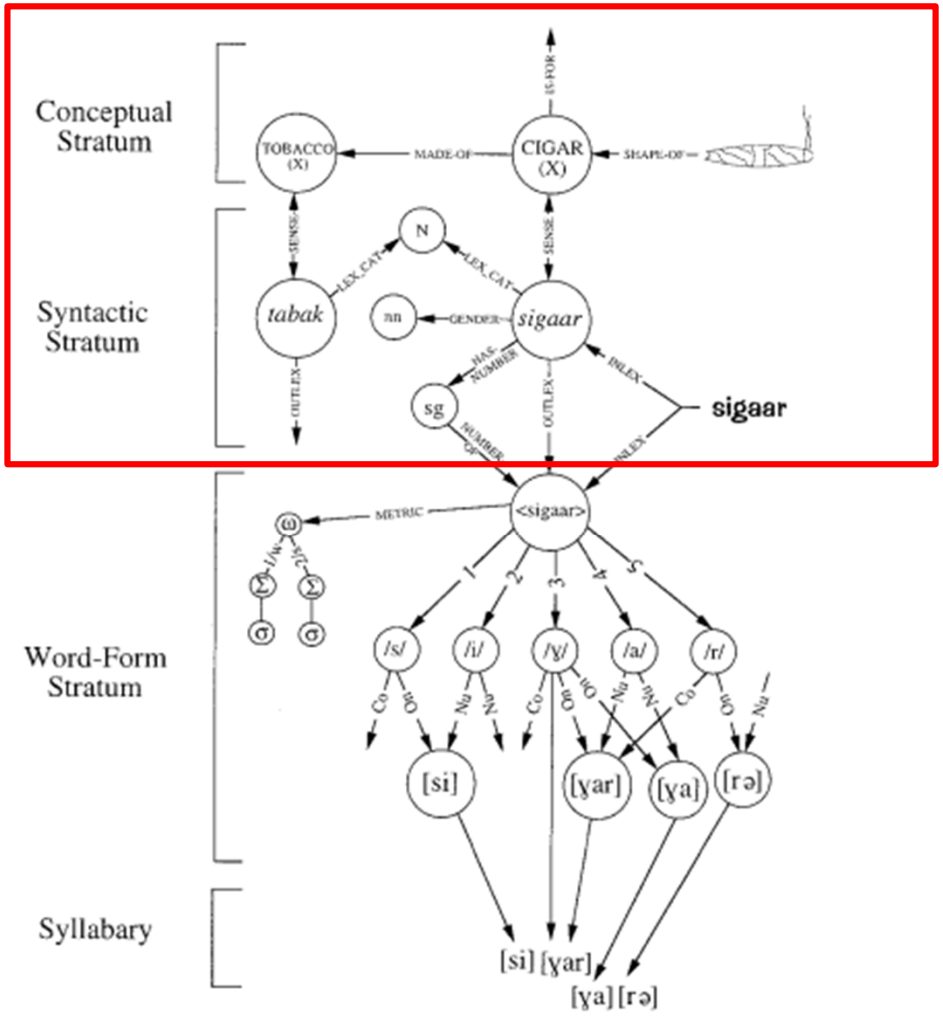


Figure 4.4. Representation of the Dutch word *sigaar* plus part of the word *tabak* in WEAVER++.

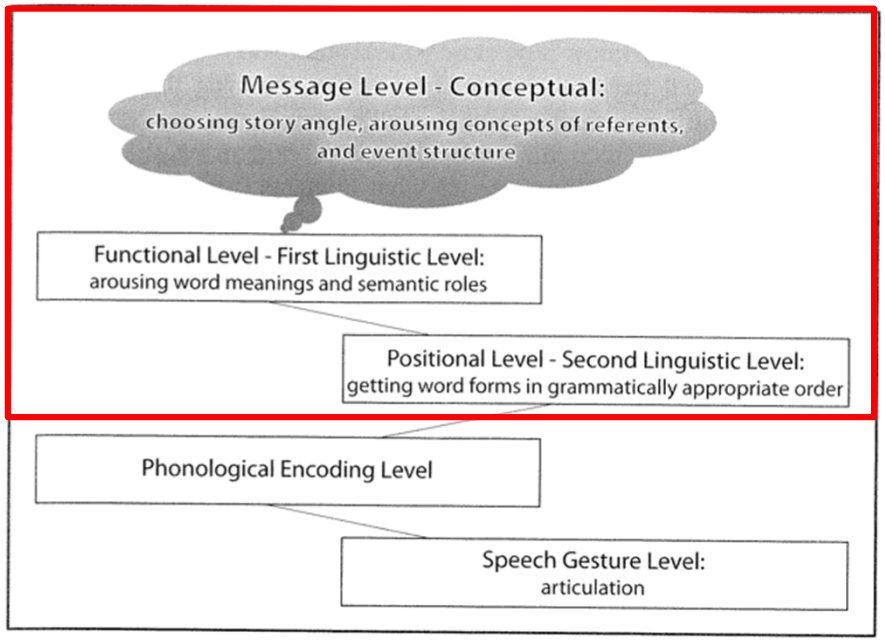


Figure 3.1. Levels of processing discussed in Chapters 3 and 4.

NB, Because the Roelofs (WEAVER++) model is focused on activation for single words, it doesn't align exactly with the more general model. E.g., the syntactic information is not "positional," per se, but single-word things like number, gender, etc.

Belke et al.

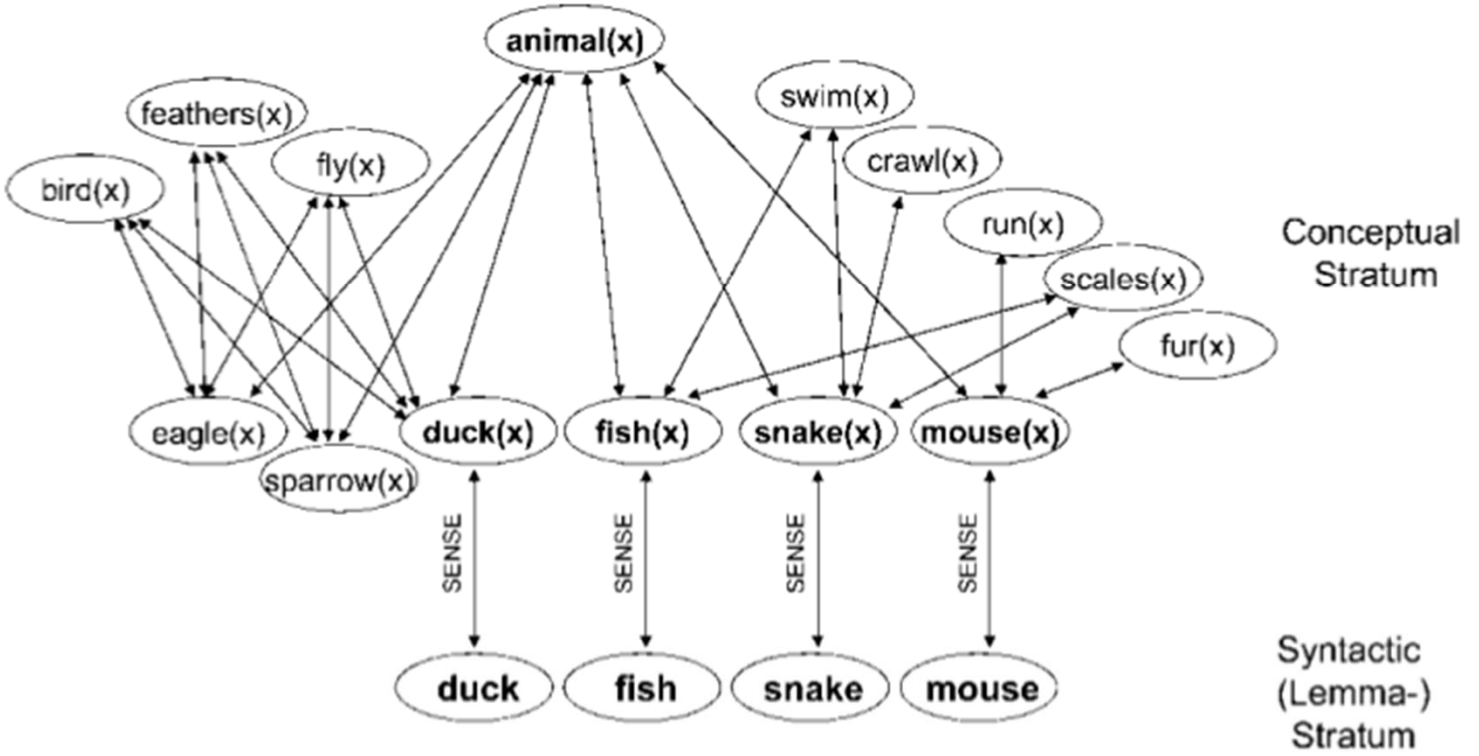
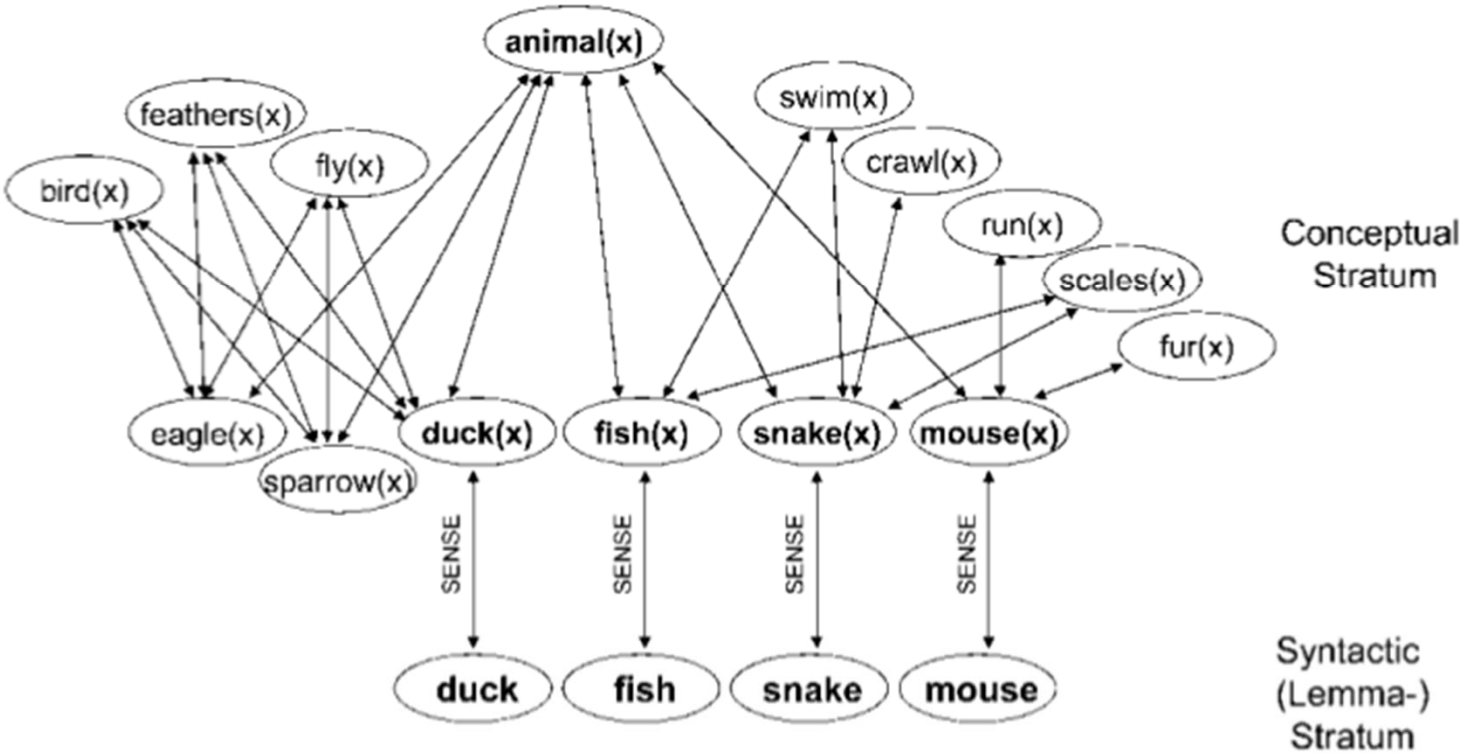


Figure 1. Fragment of the conceptual and syntactic strata, as assumed in WEAVER++ (adapted from Roelofs, 1992, and Levelt et al., 1999). Note that the arrows represent the flow of activation in the network rather than types of connection. Different types of relations exist between lexical concepts and their feature nodes, such as *is-a*, *can* and *has*. For the sake of readability, not all connections among related nodes are presented.

If we just take a naïve view of spreading activation, we might imagine that a whole block of similar pictures presented together would lead to *faster* naming times…

This model helps us understand how spreading activation can lead to *slower* naming times in some cases.

Belke et al.



In speech production, we have to distinguish carefully between *activation* and *selection.*

Spreading activation from, e.g., *duck* to *fish* means that activating one can help activate the other. But this can cause problems when our task is to *select* the correct lemma from a bunch of active ones…

The WEAVER++ model assumes that selection time is determined by a ratio of the activity of the *most active* lemma and the *overall activation of all lemmas.*

Thus, when there's feedback from the category level, this slows down lemma selection because of all the collateral activation for items from the same category.

Belke et al.

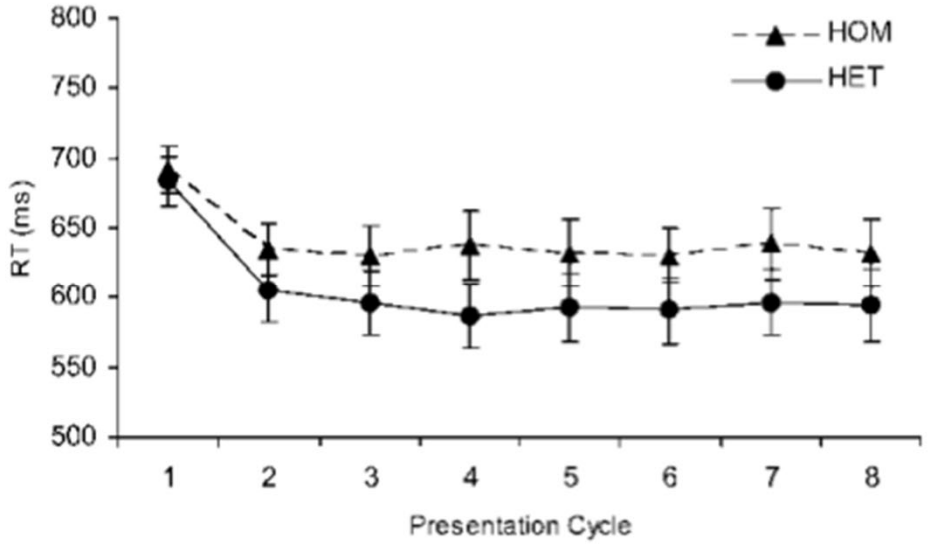


Figure 2. Experiment 1: Mean response times (RT) by context (homogeneous, hom vs. heterogeneous, het) and presentation cycle. The error bars represent one standard error.

The experiment was run for 8 "cycles." In each cycle, all of the stimuli were presented.

HOM = homogeneous blocks, e.g., "duck", "mouse", "fish", "snake"
HET = heterogeneous blocks, e.g, "duck", "saw", "chair", "bus"

The same stimulus (e.g., "duck") was presented in HOM and HET blocks in alternating cycles.

Belke et al.

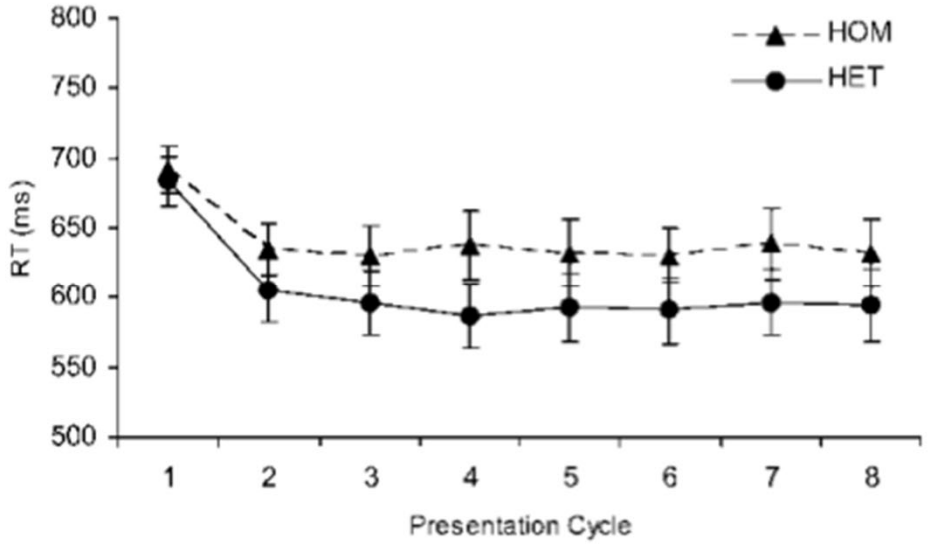


Figure 2. Experiment 1: Mean response times (RT) by context (homogeneous, hom vs. heterogeneous, het) and presentation cycle. The error bars represent one standard error.

There are actually two things happening here.

- a big identity priming effect

- a semantic blocking effect

A criticism sometimes raised against the semantic blocking paradigm as a means of studying lexical-semantic processes is that the blocking effects may arise earlier, during the visual-conceptual analysis of the items, rather than during lexical processing. One might argue that it is perhaps more difficult to discriminate between items belonging to the same semantic category, which are likely to share visual features, than between items belonging to different semantic categories. As noted above, the items used in the present experiment were selected to be as dissimilar as possible. For 12 of the 16 items we used, Damian et al. (2001) collected similarity ratings, which revealed that the items in homogeneous sets were, on average, not judged to be more visually similar than those in heterogeneous sets. In addition, the absence of a blocking effect during the first presentation of the materials argues against the allocation of the blocking effect at the visual-conceptual level: It is difficult to see why an effect of visual similarity would be absent at first and arise after the first presentation of the materials. Instead, one might expect to see the opposite pattern—visual similarity among items should be more influential at the beginning of a block, when the participants are uncertain which features discriminate between the objects, than after they have become more familiar with the objects. For a supplementary experiment with eight participants we selected items that were semantically as well as visually highly similar objects (e.g., shirt, blouse, coat, jumper; see Appendix B). We only included four presentation cycles because Experiment 1 of the present study had shown that blocking effects emerge after the first presentation cycle and remain stable thereafter. Using the visually similar items we found a context effect in the first as well as all following repetitions, resulting in significant main effects of context, $F(1, 7) = 15.24$, $MSE = 3{,}335$, $p < .01$, and presentation cycle, $F(3, 21) = 24.93$, $MSE = 2{,}075$, $p < .001$, but no significant interaction of context and presentation cycle. These findings support our assumption that semantically and visually similar sets should produce a different pattern of blocking effects across presentation cycles than should semantically similar, visually dissimilar sets.

Dealing with potential confounds 101

What if the blocking effect arises at the level of visual object recognition?

Two responses:

1. Stimuli were normed so that objects in the same category were not more visually similar to one another than objects in different categories, on average.

2. A control experiment was run in which visual similarity was blocked along, and *different results were found.*

Specifically, the blocking effect was observed in *the first block of trials*.

Thus, the effects in Experiment 1 aren't likely to be due to visual similarity.

Yeah, but what about that Hermens et al., study, where people saw multiple objects simultaneously, and they didn't find an effect?

Maybe there's just something weird about seeing objects in serial?



Belke et al. address this by asking participants to view displays of four items simultaneously and produce their names in serial.

They measure eye-movements while they do this.

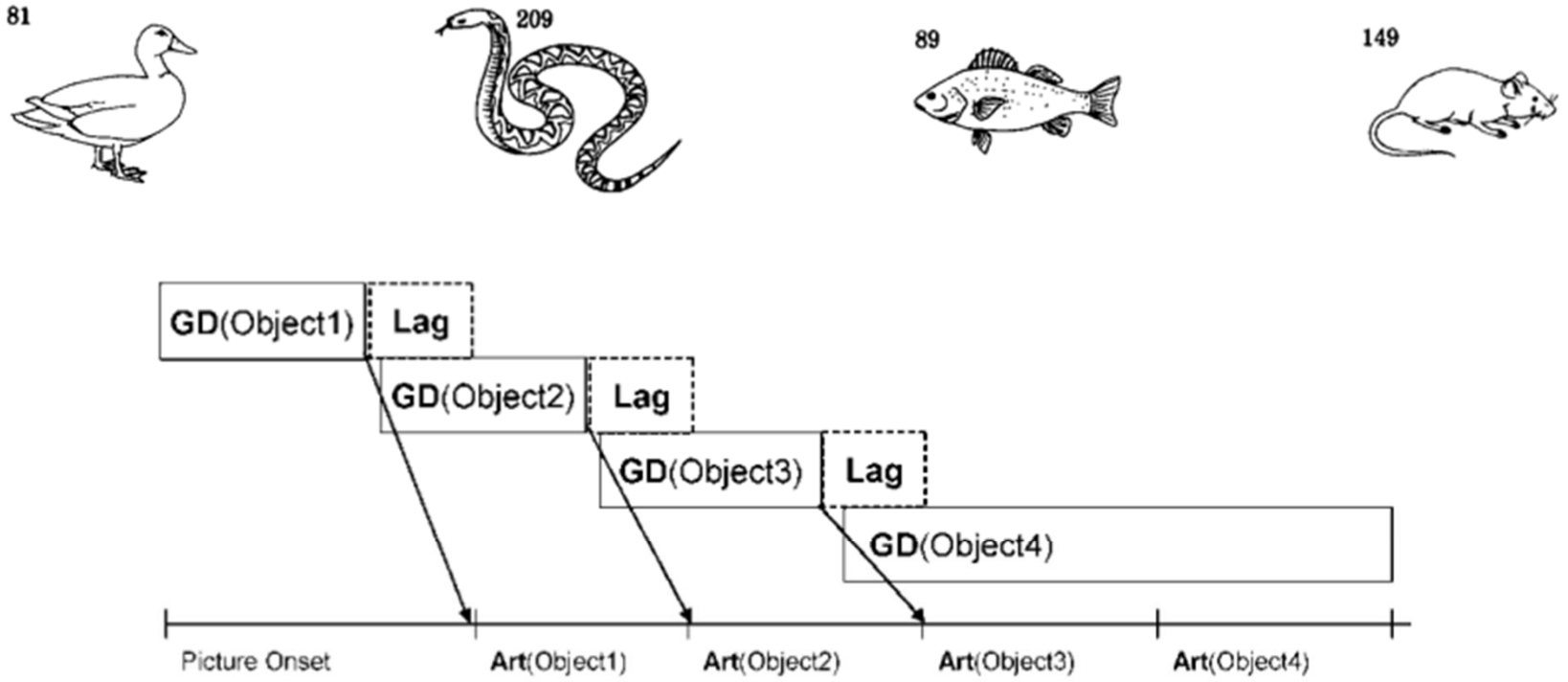


Figure 3. Schematic overview of the temporal coordination of gaze durations (GD) and articulatory word onsets (Art) in a multiple-object naming task.

Note there are two key *dependent variables* here:

gaze duration – there's evidence that lemma selection is coordinated with eye-movements

lag – "downstream" processes like phonological encoding and gesture planning can happen without visual support, once the eyes have moved on.

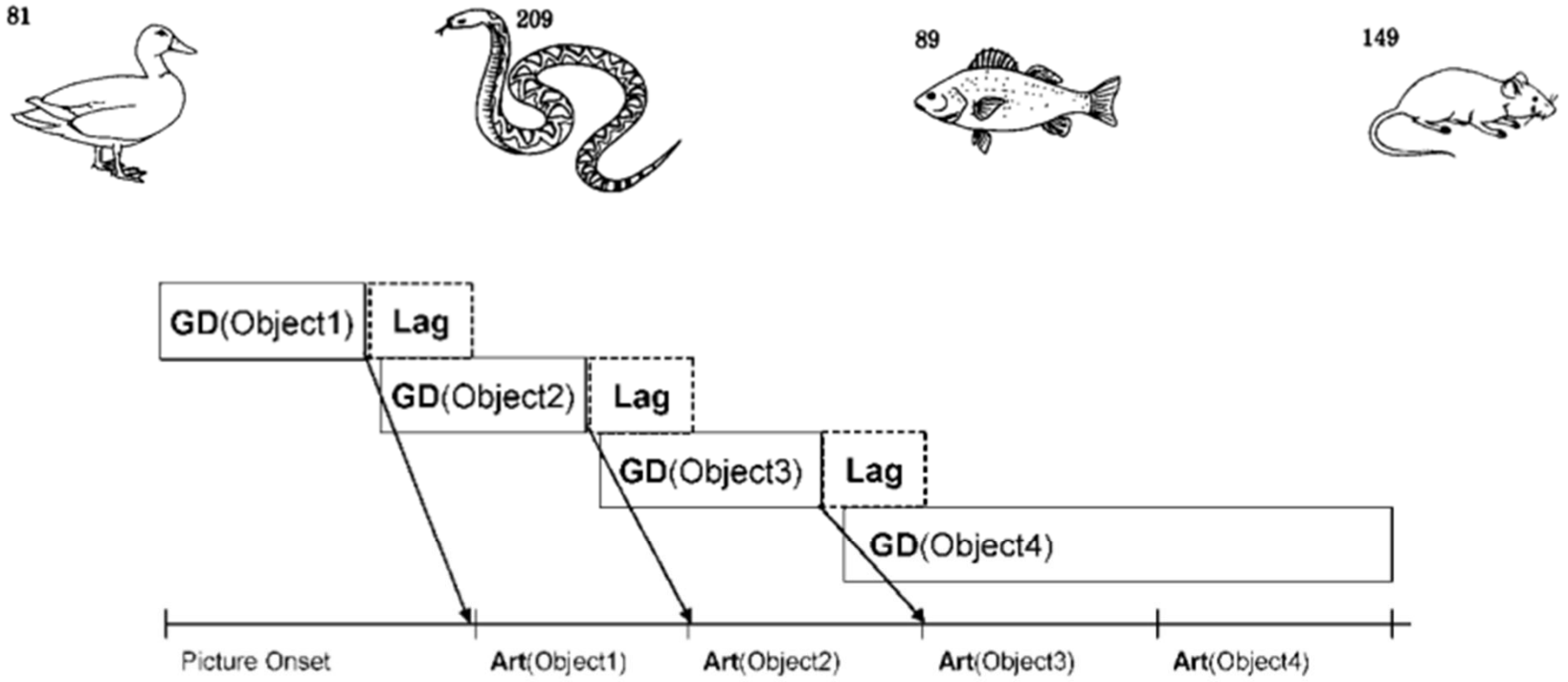
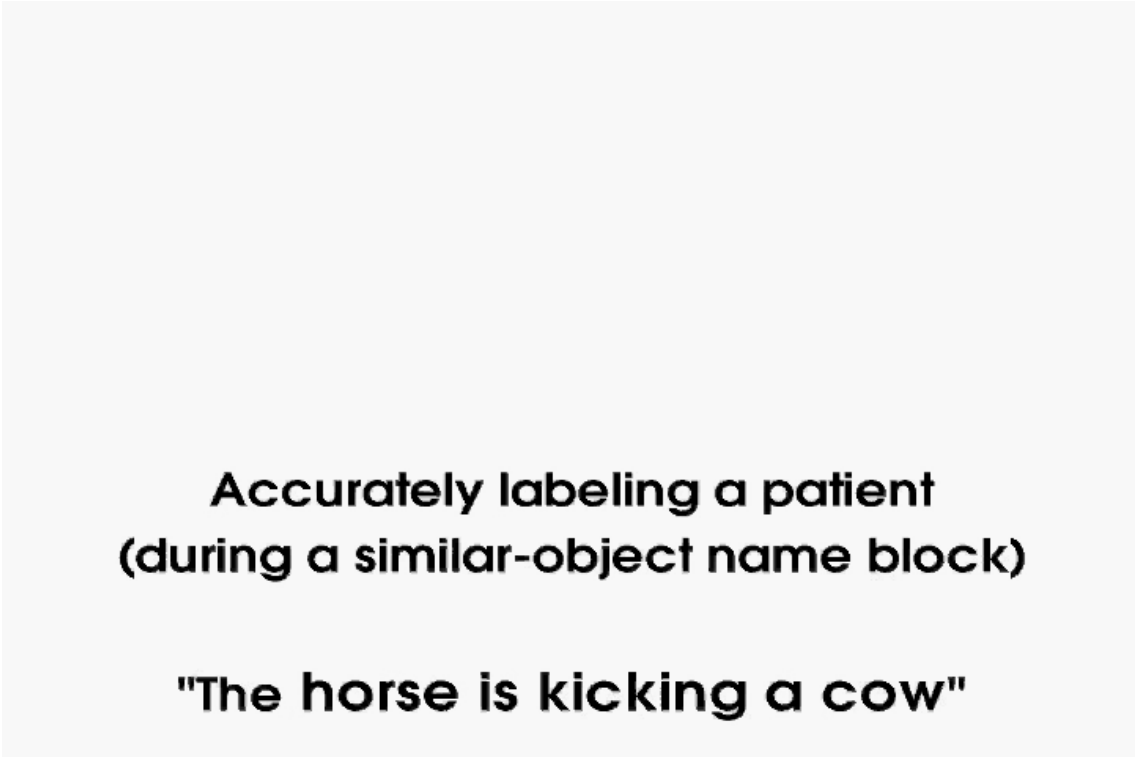


Figure 3. Schematic overview of the temporal coordination of gaze durations (GD) and articulatory word onsets (Art) in a multiple-object naming task.

What eye-mouth coordination looks like (from Griffin, Bock, and colleagues)



nice demos from: http://homepage.psy.utexas.edu/homepage/faculty/griffin/Demos.html

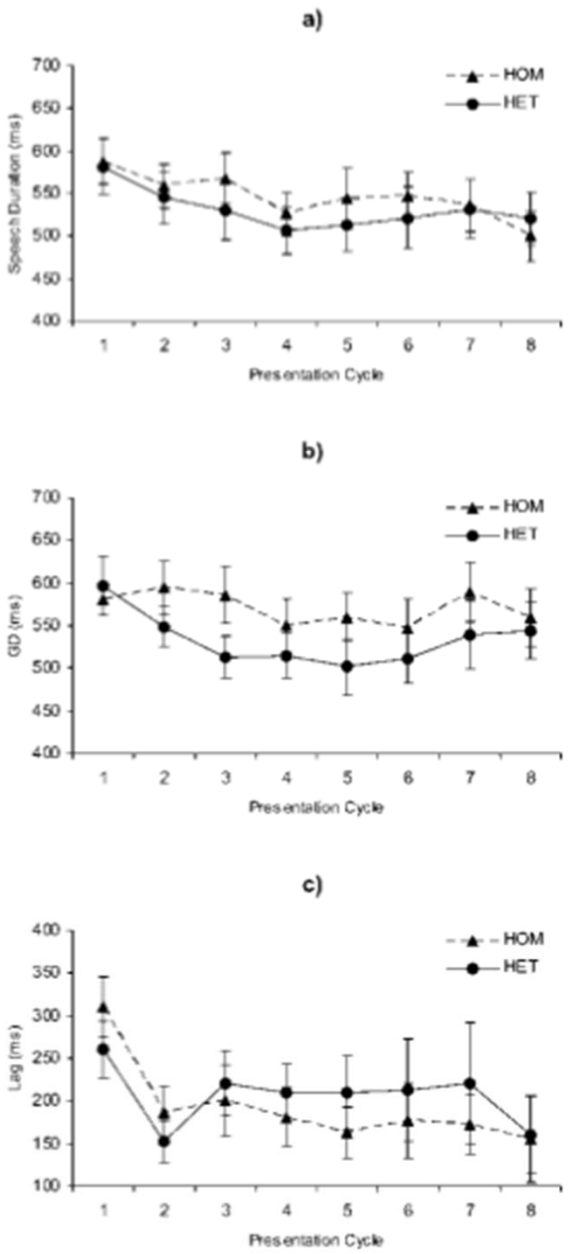


Figure 4. Experiment 2: (a) Mean speech durations, (b) gaze durations (GD), and (c) eye–speech lags by context and presentation cycle. The error bars represent one standard error.

Overall duration of speech decreases over the course of the experiment, but doesn't show a blocking effect.

Gaze duration shows a blocking effect, but only after the first cycle.

There's no effect on lag. Numerically, the blocking effect seems to go in the opposite direction, but this is not significant.

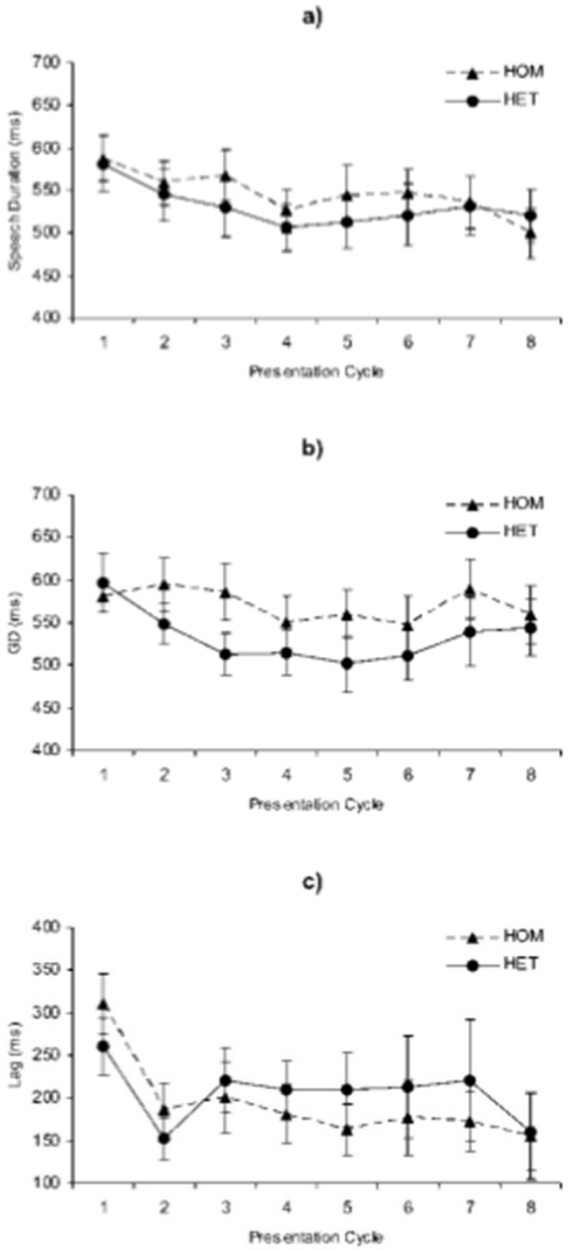


Figure 4. Experiment 2: (a) Mean speech durations, (b) gaze durations (GD), and (c) eye–speech lags by context and presentation cycle. The error bars represent one standard error.

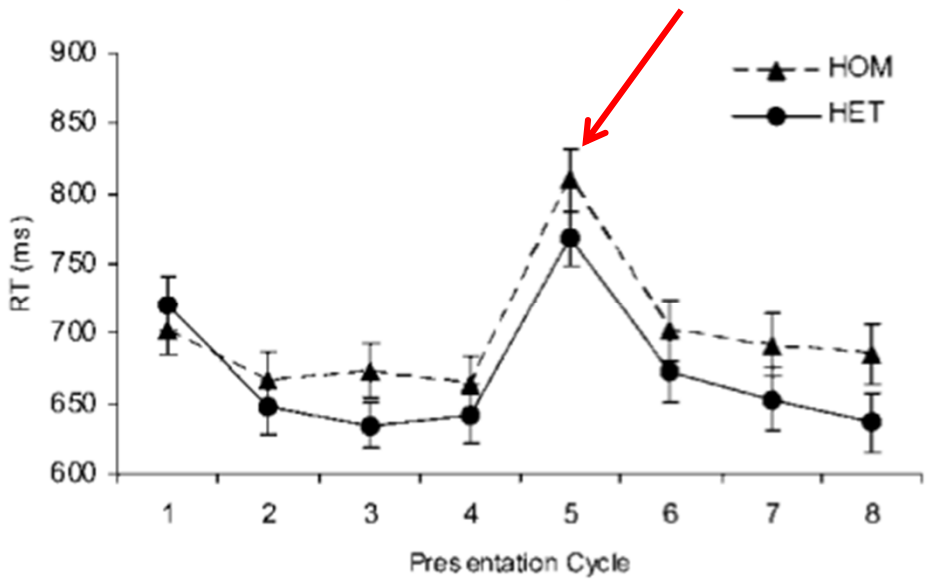
Overall duration of speech decreases over the course of the experiment, but doesn't show a blocking effect.
→*gesture level*

Gaze duration shows a blocking effect, but only after the first cycle.
→*lemma selection*

There's no effect on lag. Numerically, the blocking effect seems to go in the opposite direction, but this is not significant.
→*phonological encoding*

Experiment 3
Does blocking generalize to new items in the same category?

New items introduced



Note that the repetition priming effect goes away in Cycle 5…

Overall, performances slows down dramatically, but the blocking effect is still there.

Compare the effect on new items from a blocked category to the items in Cycle 1…

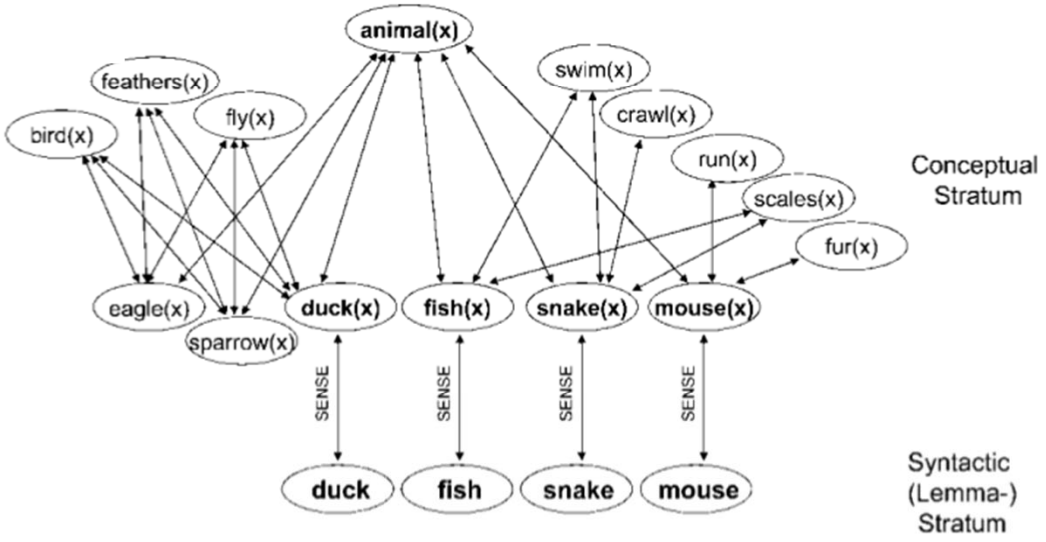


Figure 1. Fragment of the conceptual and syntactic strata, as assumed in WEAVER++ (adapted from Roelofs, 1992, and Levelt et al., 1999). Note that the arrows represent the flow of activation in the network rather than types of connection. Different types of relations exist between lexical concepts and their feature nodes, such as *is-a*, *can* and *has*. For the sake of readability, not all connections among related nodes are presented.

How do Belke et al. isolate this effect in the model:

- not visual recognition

- not phonological encoding (since words aren't similar)

- not item-specific (generalizes to new items)

…

How do Belke et al. isolate the blocking effect in the model:

- not visual recognition
  - normed/matched stimuli
  - control experiment

- not phonological encoding (?)
  - words are not phonologically similar
  - gaze duration, not eye-speech lag is impacted

- not just a specific effect of repeating lists
  - generalizes to new items

How to have an idea for an experiment

Ask: “Couldn’t it just be … ?”

For stimulus manipulations, many features of words are correlated with frequency:

- length
- age of acquisition
- concreteness/imageability
- “contextual diversity”
- etc.

So, what if frequency effects are just a catch-all for these other phenomena?

For experimental manipulations, consider the many different things that go on even in this very simple blocking study:

- object recognition
- response selection

For next session:

Read:
Diehl, Lotto, & Holt (2004, ***especially*** *Intro and pp. 155-159)*